Model Guide: Bayesian Forecasting of Household Electricity Demand

The model, every equation and symbol, and what the results mean

Project 17

October 10, 2026

This guide collects the model-related explanations in one place: the data that enter the model, the equations and the meaning of every symbol, the priors, how the posterior was computed and checked, how forecasts are made and scored, what the results say, the limitations, and the recommended next steps. Full derivations and reproducible numbers are in `report.tex`, `src/model.py`, and `results/`.

1 What is being modelled

- The data come from one household near Paris, recorded from December 2006 through November 2010.
- The raw file records the average active power in every minute, in kilowatts (kW).
- A minute is 1/60 of an hour, so the energy used on day d is

$$y_d = \frac{1}{60} \sum_{m \in d} \mathrm{GAP}_m \qquad [\mathrm{kWh}]. \tag{1}$$

Symbol	Meaning
d	A calendar day.
$m \in d$	The recorded minutes m that belong to day d.
GAP_m	Global active power in minute m, in kW.
$\sum_{m \in d}$	Add up the values over all recorded minutes of day d.
$1/60$	Converts one minute at a power of 1 kW into kWh. Equation (1) is a daily *sum*, not a daily mean.
y_d	Total energy used on day d, in kilowatt-hours (kWh).

- The two partial days at the start and end of the record are dropped. A remaining day counts as observed only if at least 95% of its 1,440 minutes are present.
- The result is a gap-free grid of $T = 1{,}440$ calendar days: 1,417 observed and 23 missing. Missing days stay on the grid, so the day-to-day error process remains continuous.

The scientific goals are to describe long-term, yearly, and weekly patterns; to forecast future daily demand with honest uncertainty; and to identify unusual days or periods.

2 The general model

Every model fitted in the project belongs to this family:

$$z_t = g(y_t), \tag{2}$$

$$z_t = \mathbf{x}_t^\top \boldsymbol{\beta} + \varepsilon_t, \tag{3}$$

$$\mathbf{x}_t^\top \boldsymbol{\beta} = \beta_0 + \beta_1 \frac{t - \bar{t}}{365} + \sum_{j=1}^{J} \left[a_j \sin\left(\frac{2\pi j t}{365}\right) + b_j \cos\left(\frac{2\pi j t}{365}\right) \right] + \sum_{k=2}^{7} \delta_k \, \mathbf{1}\{\text{dow}_t = k\}, \tag{4}$$

$$\varepsilon_t = \sum_{\ell=1}^{p} \phi_\ell \, \varepsilon_{t-\ell} + \eta_t, \qquad \eta_t \sim t_\nu(0, \sigma^2). \tag{5}$$

OR: $\eta_t \mid \lambda_t \sim N(0, \sigma^2/\lambda_t)$?

In words: daily consumption = a calendar-based prediction + a leftover that partly carries over from day to day, with occasional very large surprises allowed. Equation (3) splits each day into the predictable part and the leftover, equation (4) is the predictable part, and equation (5) describes the leftover.

2.1 How to read the notation

Notation	Meaning
$=$	Equals.
$+$, $-$	Add, subtract.
a/b or $\frac{a}{b}$	a divided by b.
Subscript, e.g. y_t	Identifies which day, lag, harmonic, or category the quantity belongs to.
Superscript 2, e.g. σ^2	The quantity squared.
Bold, e.g. $\mathbf{x}$, $\boldsymbol{\beta}$	A vector: a list of several numbers handled together.
$\sum_{j=1}^{J}$	Add the terms for $j = 1, 2, \ldots, J$.
$\in$	"Belongs to", e.g. a minute belonging to a day.
$\approx$	Approximately equal to.
$<$, $>$, $\leq$, $\geq$	Less than, greater than, at most, at least.
$\sim$	"Is distributed as": the quantity on the left follows the probability distribution on the right.
$\mid$	"Given", or "conditional on".
$P(A \mid B)$	Probability of A, given the information B.
$\text{Var}(X)$, $\text{SD}(X)$	Variance and standard deviation of X.
$[L, U]$ after an estimate	Lower and upper ends of an interval. For parameters in this guide, a 95% posterior credible interval.

2.2 Response and time

Symbol	Meaning
t	Day number on the gap-free calendar, $t = 1, \ldots, T$.
T	Number of calendar days, $T = 1{,}440$.
y_t	Electricity used on day t, in kWh. Observed on 1,417 days; imputed by the model on the other 23.
$g(\cdot)$	Transformation applied to y_t before modelling.
z_t	The response on the modelling scale. Identity model: $z_t = y_t$. Log model: $z_t = \log y_t$.
$\bar{t}$	The average day index. Centring time at $\bar{t}$ makes the trend term zero at the middle of the record and reduces the correlation between the trend and the intercept.

The headline model M7 uses the identity transformation, so $z_t = y_t$ and all of its coefficients are in kWh.

2.3 The calendar prediction

Symbol	Meaning
$\mathbf{x}_t$	The known calendar inputs for day t: a constant 1, centred time, the sine and cosine terms, and six weekday indicators.
$\boldsymbol{\beta}$	All regression coefficients collected into one vector.
$\top$	Transpose. $\mathbf{x}_t^\top \boldsymbol{\beta}$ multiplies each input by its coefficient and adds the results, giving the calendar prediction for day t.
β_0	Intercept: expected use on a Monday at the middle of the record, before the seasonal waves are added (the waves average to zero over a year). It is not, by itself, the average-day level.
β_1	Trend: change in daily use per year, because $(t - \bar{t})/365$ measures time in years.
365	Length of the yearly cycle in days, as required by the project brief.

The reported *average-day level at the record midpoint* adds the average weekday offset to the intercept. For M7 it is 25.96 kWh, while the Monday-reference intercept itself has posterior mean 23.82 kWh.

2.4 Annual harmonics

Symbol	Meaning
J	Number of harmonic pairs (one sine plus one cosine each).
j	Harmonic number, $j = 1, \ldots, J$.
$j = 1$	One full wave every 365 days: $\sin(2\pi t/365)$ and $\cos(2\pi t/365)$.
$j = 2$	Two waves every 365 days (one every 182.5 days): $\sin(4\pi t/365)$ and $\cos(4\pi t/365)$. Added to the first harmonic, it makes the yearly curve asymmetric.
a_j	Coefficient on the sine term of harmonic j, in kWh.
b_j	Coefficient on the cosine term of harmonic j, in kWh.
π	The constant 3.14159...
$\sin(\cdot)$, $\cos(\cdot)$	Smooth periodic functions. Dividing $2\pi jt$ by 365 makes harmonic j repeat j times a year.

Each sine–cosine pair can be rewritten as a single wave with a size and a timing:

$$a_j \sin(\omega_j t) + b_j \cos(\omega_j t) = A_j \cos(\omega_j t - \psi_j), \tag{6}$$

$$\omega_j = \frac{2\pi j}{365}, \qquad A_j = \sqrt{a_j^2 + b_j^2}, \qquad \psi_j = \text{atan2}(a_j,\, b_j). \tag{7}$$

Symbol	Meaning
ω_j	Angular frequency of harmonic j, in radians per day.
A_j	Amplitude: how far that wave rises above (and falls below) zero, in kWh.
$2A_j$	Peak-to-trough swing of that wave on its own.
ψ_j	Phase: sets the date of the wave's peak.
atan2(a, b)	The angle whose sine and cosine are proportional to a and b; using the signs of both puts it in the correct quadrant.
$\sqrt{\cdot}$	Square root.

The full seasonal curve is the sum of both waves, so its total range is generally not $2A_1 + 2A_2$.

2.5 Weekday effects

Symbol	Meaning
dow_t	Day of the week of day t (1 = Monday, ..., 7 = Sunday).
k	A weekday from Tuesday ($k = 2$) to Sunday ($k = 7$).
δ_k	How much weekday k differs from Monday, in kWh.
$\mathbf{1}\{\text{dow}_t = k\}$	Indicator: equals 1 if day t is weekday k, and 0 otherwise.

Monday has no coefficient of its own; it is the reference day. For reporting, the seven weekday values are re-centred on their weekly mean, which answers the more natural question: how far is this weekday above or below an average day?

2.6 The leftover: an autoregressive error

Symbol	Meaning
ε_t	Residual (leftover) on day t: actual use minus the calendar prediction.
p	Autoregressive order: how many previous days' residuals feed into today.
ℓ	Lag, $\ell = 1, \ldots, p$: how many days back.
ϕ_ℓ	Persistence: the fraction of the residual from ℓ days ago that carries into today.
η_t	Innovation: the genuinely new surprise on day t, after the carry-over from earlier days is accounted for.

For M7, $p = 1$, so the error equation is

$$\varepsilon_t = \phi\, \varepsilon_{t-1} + \eta_t. \tag{8}$$

Part of yesterday's unexplained high or low use carries into today, and then a new surprise is added. The fitted ϕ is about 0.375, so the memory fades within a few days.

2.7 Heavy-tailed surprises

Each new surprise follows a Student-t distribution rather than a normal one:

$$\eta_t \sim t_\nu(0,\, \sigma^2). \tag{9}$$

Symbol	Meaning
$t_\nu(0, \sigma^2)$	Student-t distribution centred at 0, with scale σ and ν degrees of freedom.
σ	Innovation scale: the typical size of a new one-day surprise, in kWh.
σ^2	The scale squared.
ν	Degrees of freedom: controls how heavy the tails are. A small ν allows more extreme days; as ν grows, the distribution approaches a normal distribution.

Strictly, σ is a *scale*, not the standard deviation. When $\nu > 2$,

$$\mathrm{SD}(\eta_t) = \sigma \sqrt{\frac{\nu}{\nu - 2}}. \tag{10}$$

The project output labels σ the "one-step innovation sd"; "innovation scale" is the precise name.

The sampler writes the Student-t as a normal distribution whose variance changes from day to day:

$$\eta_t \mid \lambda_t \sim N\left(0,\ \frac{\sigma^2}{\lambda_t}\right), \qquad \lambda_t \sim \text{Gamma}\left(\frac{\nu}{2},\ \frac{\nu}{2}\right). \tag{11}$$

Symbol	Meaning
$N(\mu,\, v)$	Normal distribution with mean μ and variance v.
λ_t	Hidden weight for day t. Near 1 on an ordinary day; small on an extreme day.
σ^2/λ_t	Variance of day t's surprise. A small λ_t makes it large, so an unusual day is absorbed without inflating the uncertainty of every ordinary day.
Gamma$(\alpha,\, \beta)$	Gamma distribution with shape α and rate β. With $\alpha = \beta = \nu/2$ the weights average exactly 1.

Averaging over λ_t gives back the Student-t distribution in equation (9).

2.8 Missing days

The 23 missing values of y_t are neither set to zero nor deleted. Their z_t and ε_t are sampled from the model along with everything else. This keeps the day-to-day error link intact across gaps and carries the uncertainty about those days into the parameter estimates and the forecasts.

3 The headline model, M7

The model behind the main conclusions is

$$y_t = \mathbf{x}_t^\top \boldsymbol{\beta} + \varepsilon_t, \tag{12}$$

$$\begin{aligned}\mathbf{x}_t^\top \boldsymbol{\beta} = \beta_0 + \beta_1 \frac{t - \bar{t}}{365} + a_1 \sin\left(\frac{2\pi t}{365}\right) + b_1 \cos\left(\frac{2\pi t}{365}\right) \\ + a_2 \sin\left(\frac{4\pi t}{365}\right) + b_2 \cos\left(\frac{4\pi t}{365}\right) + \sum_{k=2}^{7} \delta_k \, \mathbf{1}\{\text{dow}_t = k\},\end{aligned} \tag{13}$$

$$\varepsilon_t = \phi \, \varepsilon_{t-1} + \eta_t, \qquad \eta_t \sim t_\nu(0, \sigma^2). \tag{14}$$

In short: the response is measured directly in kWh, and the model has a centred linear trend, $J = 2$ annual harmonic pairs, six weekday indicators with Monday as the reference, an AR(1) residual, and Student-t innovations.

M7 was chosen as the headline model because it was the best of the seven candidates on the held-out 12-month forecast, and its predictive intervals were much sharper and better calibrated than those of the log-scale versions. It is the best model in this candidate set, not a claim that no better model exists.

4 Priors

Because the model is Bayesian, every unknown parameter gets a probability distribution before the data are seen. For M7:

$$\beta_0 \sim N(0,\ 100^2), \tag{15}$$

$$\beta_r \sim N(0,\ 40^2) \quad \text{for every other regression coefficient } r, \tag{16}$$

$$\sigma \sim \text{half-}t_3(0,\ 20), \tag{17}$$

$$\phi \sim \text{Uniform}(-1,\ 1), \tag{18}$$

$$\nu - 2 \sim \text{Exponential(mean 10)}. \tag{19}$$

Prior	**Meaning**
$N(0,\ 100^2)$ and $N(0,\ 40^2)$	Normal distributions centred at 0 with standard deviations 100 and 40 kWh. They are deliberately wide, so the data, not the prior, determine the coefficients.
half-$t_3(0,\ 20)$	A Student-t with 3 degrees of freedom and scale 20, folded at zero. It keeps $\sigma > 0$ while allowing a broad range of values.
Uniform$(-1,\ 1)$	Every value of ϕ between -1 and 1 is equally likely beforehand. $-1 < \phi < 1$ is the stationary region, where shocks fade instead of growing without bound.
Exponential(mean 10) on $\nu - 2$	Forces $\nu > 2$, so the variance of the Student-t exists, while still allowing heavy tails.
Missing y_t	No separate prior: their distribution is implied by the regression and the AR process.

The log-scale candidates use smaller prior scales because their coefficients are much smaller: 5 for the intercept, 2 for the other coefficients, and 1 for the half-t scale.

5 Candidate models and why they were compared

Model	***J***	**Error process**	**Scale**	**What the comparison tests**
M1	1	independent, Gaussian	log	naive baseline
M2	1	AR(1), Student-t	log	day-to-day memory and heavy tails
M3	2	AR(1), Student-t	log	the required second harmonic
M4	3	AR(1), Student-t	log	a more flexible yearly curve
M5	2	AR(2), Student-t	log	longer residual memory
M6	2	AR(1), Gaussian	log	the value of heavy tails
M7	2	AR(1), Student-t	kWh	whether the log transform helps

WAIC, an in-sample estimate of predictive accuracy, preferred M5. The held-out forecasts rejected AR(2): the extra lag was fitting long absence blocks that did not generalise. The held-out year also showed that modelling kWh directly beat the log transform. Forecast performance, not WAIC alone, therefore decided the headline model.

Adding a third harmonic improved the log-scale model, but a three-harmonic model on the kWh scale was not fitted. That remains an important extension.

6 How the posterior was computed and checked

The posterior distribution was computed with a purpose-built Markov chain Monte Carlo (MCMC) sampler:

- Gibbs updates for the regression coefficients, the innovation scale, the hidden weights λ_t, and the missing responses;
- random-walk Metropolis updates for the AR coefficients and for $\log(\nu - 2)$;
- exact stationary initialisation of the first AR residuals;
- four chains of 40,000 iterations for the full-data fits, with the first 10,000 of each discarded as burn-in and every fifth remaining iteration kept, giving 24,000 posterior draws.

Check	Meaning and result
$\widehat{R}$	Compares the spread within chains with the spread between chains. Values near 1 (target below 1.01) indicate convergence. M7's worst value: 1.0069.
ESS	Effective sample size: how many independent draws the correlated MCMC draws are worth (target above 400). M7's smallest bulk ESS: 1,072.
Simulation check	On data simulated with known parameters, 14 of 15 parameters fell inside their 95% credible intervals, as expected.

7 Forecast equations

For one posterior draw, the forecast for h days after the origin T is built as

$$\mu_{T+h} = \mathbf{x}_{T+h}^{\top}\boldsymbol{\beta}, \tag{20}$$

$$\varepsilon_{T+h} = \phi\,\varepsilon_{T+h-1} + \eta_{T+h}, \tag{21}$$

$$y_{T+h} = \mu_{T+h} + \varepsilon_{T+h}. \tag{22}$$

Symbol	Meaning
T	Forecast origin: the last day whose data are used.
h	Horizon: how many days ahead, $h = 1, 2, \ldots$
μ_{T+h}	Calendar prediction for day $T + h$.
ε_{T+h}	Simulated future residual, built forward one day at a time from the last known residual.
y_{T+h}	Simulated future consumption.

Repeating this for every posterior draw gives the posterior predictive distribution. It includes the uncertainty in the coefficients and in ϕ, σ, and ν, the future innovations, and their propagation through the AR process.

With the parameters held fixed and an innovation variance v_η, the forecast variance of the residual h days ahead is

$$\operatorname{Var}(\varepsilon_{T+h} \mid \varepsilon_T) = v_\eta \sum_{r=0}^{h-1} \phi^{2r}, \qquad v_\eta = \sigma^2 \frac{\nu}{\nu - 2}. \tag{23}$$

Symbol	Meaning
v_η	Variance of one innovation η_t.
r	How many days before the target day a shock occurred.
ϕ^{2r}	Contribution of that shock to the forecast variance, relative to a same-day shock. Each day of delay multiplies a shock's effect by ϕ, and so its variance by ϕ^2.

As h grows, the forecast standard deviation relative to the one-day-ahead value approaches

$$\frac{1}{\sqrt{1 - \phi^2}}. \tag{24}$$

With $\phi \approx 0.375$ the ceiling is about 1.08. Knowing the latest residual therefore makes near-term intervals about 8% narrower, and almost all of that advantage is gone within four to seven days.

8 How the forecasts were evaluated

The final twelve months (1 December 2009 to 25 November 2010) were held out and never used to estimate the parameters.

- **Single origin:** forecast the whole held-out year from 30 November 2009.
- **Rolling origin:** forecast 1 to 60 days ahead from many starting days in the held-out year, each time updating the current residual with the newly observed consumption.

Score	Meaning
Log predictive density (LPD)	Rewards a forecast distribution that is both sharp and centred in the right place. Higher is better.
CRPS	Continuous ranked probability score: forecast error that uses the whole forecast distribution, in kWh. Lower is better.
Coverage	Share of actual values inside the nominal 50% or 95% predictive interval. Good calibration means coverage close to the nominal percentage.
PIT	Probability integral transform: the forecast's cumulative probability at the actual value. Roughly uniform PIT values indicate good calibration.

M7's rolling 95% intervals covered 94.5% of actual values one day ahead and 95.2% seven days ahead. Their average width rose from about 28.9 kWh one day ahead to about 31.4 kWh seven days ahead and then stayed almost flat.

9 Main findings from M7

9.1 Overall level and trend

All intervals in this section are 95% posterior credible intervals.

Quantity	Posterior mean [95% interval]
Average-day level at the record midpoint	25.96 [25.41, 26.50] kWh/day
Trend	−0.186 [−0.667, 0.292] kWh/year
Trend as a percentage of the level	−0.72 [−2.57, 1.13] % per year
$P(\text{slope} < 0 \mid \text{data})$	0.775

The interval for the trend includes zero, so there is no conclusive long-term rise or fall.

9.2 Yearly pattern

Quantity	Posterior mean [95% interval]
First-harmonic amplitude A_1	8.08 [7.31, 8.85] kWh
First-harmonic peak-to-trough swing $2A_1$	16.15 [14.62, 17.71] kWh
First-harmonic peak date	about 14 January [9 January, 20 January]
Second-harmonic amplitude A_2	2.43 [1.68, 3.19] kWh
Amplitude ratio A_2/A_1	0.302 [0.204, 0.405]
$P(A_2 > 0.15\,A_1 \mid \text{data})$	0.999
Full seasonal range (maximum minus minimum)	18.25 [16.48, 20.06] kWh

The second harmonic clearly matters: it sharpens the winter peak and makes the summer low longer and flatter.

9.3 Weekly pattern

Deviations from the weekly mean, in kWh per day:

Day	Posterior mean [95% interval]
Monday	−2.15 [−2.91, −1.40]
Tuesday	−0.41 [−1.17, 0.37]
Wednesday	−0.10 [−0.89, 0.69]
Thursday	−2.75 [−3.51, −1.99]
Friday	−1.12 [−1.88, −0.36]
Saturday	+3.50 [2.67, 4.33]
Sunday	+3.02 [2.21, 3.85]
Weekend minus weekday	4.56 [3.71, 5.42]
$P(\text{weekend} > \text{weekday} \mid \text{data})$	1.0000

9.4 Error process

Quantity	Posterior mean [95% interval]
Innovation scale σ	5.59 [5.25, 5.94] kWh
AR coefficient ϕ	0.375 [0.323, 0.426]
Degrees of freedom ν	5.81 [4.38, 7.81]
Long-run to one-step forecast sd ratio, $1/\sqrt{1-\phi^2}$	1.079 [1.057, 1.105]

These values show modest short-term persistence and clearly heavier tails than a normal distribution.

9.5 Unusual days and periods

The day-by-day check uses the posterior predictive p-value

$$p_t = P(y_t^{\text{rep}} \leq y_t \mid \mathbf{y}) \,. \tag{25}$$

Symbol	Meaning
y_t^{rep}	A replicate of day t simulated from the fitted model.
$\mathbf{y}$	The full observed data set.
p_t	Probability that a replicate is no larger than what actually happened. Near 0: unusually low use. Near 1: unusually high use.

- The one-day-ahead check flagged 31 days in the extreme 1% tails: 23 unusually high and 8 unusually low, against about 14 of each expected by chance.
- A one-day-ahead check can miss a sustained absence, because the AR(1) term adapts after the first day or two.
- A separate check on 14-day average consumption found 13 sustained unusual episodes.
- The clearest low periods look like summer vacancies; the high winter episodes are consistent with cold spells.

These are interpretations based on timing and load shape, not on measured occupancy or weather.

10 What the model tells us in practice

- Winter versus summer is the largest predictable difference.
- This household uses more electricity on weekends than on weekdays.
- There is not enough evidence to plan around a long-term rise or fall.
- Yesterday's consumption improves forecasts mainly for the next few days.
- Beyond about a week, the forecast is effectively a seasonal-plus-weekday distribution.
- Single-day forecasts stay wide because one household's behaviour is inherently variable.
- Plan with forecast intervals; a point forecast alone hides most of the relevant uncertainty.

11 Limitations

- This is one household, not a population or an electricity grid. The results should not be generalised without new data.
- There is no temperature variable. The harmonics capture average weather for the time of year but cannot predict a specific cold snap.
- There is no occupancy variable. Vacations are treated as outliers rather than as a separate away-from-home state.
- Weekday effects are additive and constant, although the exploratory work shows a larger weekend premium in winter.
- The innovation scale is constant, although residual variation is larger in winter than in summer.
- Trend and season are mildly confounded, because the record is just short of four complete years.
- The required 365-day period ignores leap-year drift.
- Conditional WAIC is not fully reliable for choosing forecasting models with autocorrelated data; the held-out results are more trustworthy.
- A simple day-of-year average ("climatology") beat every Bayesian candidate on the single-origin one-year forecast. Part of its advantage comes from similar August absences in the training and test years, but it also shows that a two-harmonic yearly curve is not flexible enough for the best long-range forecasts.

12 Recommended next steps

1. Add daily Paris-area temperature; this is the most valuable missing predictor.
2. Fit kWh-scale models with more harmonics, $J = 3$ through $J = 6$, since only the log-scale three-harmonic model was tested.
3. Compare versions with and without the linear trend for long-horizon forecasts.
4. Let the innovation scale vary by season, so winter can be more variable than summer.
5. Let the weekend effect depend on the season.
6. Consider a hidden home/away state for multi-week absences.
7. Include the day-of-year climatology in the same rolling-origin evaluation as the Bayesian candidates.
8. Use a rolling origin on every day, rather than every third day, and score all horizons on a common set of target dates.

13 Distinctions that are easy to mix up

Distinction	Explanation
Coefficient vs. reported effect	δ_k is measured relative to Monday; the reported weekday profile is re-centred on the weekly mean.
Residual vs. innovation	ε_t includes carry-over from earlier days; η_t is only the new one-day shock.
Credible vs. predictive interval	A credible interval describes uncertainty about a parameter; a predictive interval describes uncertainty about a future observation and is much wider.
Innovation scale vs. overall variability	σ governs one new shock; AR persistence makes longer-run residual variability larger.
In-sample fit vs. forecast skill	WAIC summarises fit to the data used for estimation; the held-out year measures forecasting directly.
Best candidate vs. best possible model	M7 is the headline model among M1–M7; the climatology result and the untested kWh models show clear room for improvement.

14 Where to verify each part

Topic	File
Core equations, sampler, forecasting, diagnostics	`src/model.py`
Candidate models and WAIC	`src/03_fit_models.py`
Held-out evaluation	`src/04_holdout.py`
Derived effects and anomaly checks	`src/05_derived.py`
Posterior summaries	`results/03_convergence.txt`
WAIC comparison	`results/03_waic.txt`
Forecast results	`results/04_holdout.txt`
Interpreted posterior quantities	`results/05_derived.txt`
Full technical narrative	`report.tex`